

Clicker Questions on Arrays #2

CS 141

What does this do?

Discuss and answer we'll go over it

```
public static int mystery(int[] arr) {  
    int x = arr[0];  
    for (int i=0; i<arr.length; i++) {  
        if(arr[i] < x) {  
            x = arr[i];  
        }  
    }  
    return x;  
}
```

Return the minimum value of an array

```
public static int mystery(int[] arr) {  
    int x = arr[0];  
    for (int i=0; i<arr.length; i++) {  
        if(arr[i] < x) {  
            x = arr[i];  
        }  
    }  
    return x;  
}
```

What if the array has multiple copies of the minimum value?

A: Now the function won't work!

not A: The function still works fine!

What if the array has multiple copies of the minimum value?

A: Now the function won't work!

not A: The function still works fine!

It doesn't matter which copy of the minimum value we return

What is the index of the minimum value of this array?

```
int[] arr=new int[] {8, 9, 1, -1, 4, 6};
```

A: 3

B: 4

C: 5

D: 0

E: -1

What is the index of the minimum value of this array?

```
int[] arr=new int[] {8, 9, 1, -1, 4, 6};
```

A: 3

-1 is the minimum value.

B: 4

C: 5

-1 is located at *index 3*

D: 0

E: -1

What does this function do?

```
public static int mystery(int[] arr) {  
    int mindex=0;  
    for (int i=0; i<arr.length; i++) {  
        if (arr[i] < arr[mindex]) {  
            mindex=i;  
        }  
    }  
    return mindex;  
}
```


Finds the *index* of the min value!

```
public static int findMinIndex(int[] arr) {  
    int minindex=0;  
    for (int i=0; i<arr.length; i++) {  
        if (arr[i] < arr[minindex]) {  
            minindex=i;  
        }  
    }  
    return minindex;  
}
```

How many of these swap the values of a and b?

```
a = b;
```

```
b = a;
```

```
a = b;
```

```
int tmp = a;
```

```
b = tmp;
```

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a = tmp;
```

```
int tmp = a;
```

```
a = b;
```

```
b = a;
```

A. 0

B. 1

C. 2

D. 3

E. 4+

How many of these swap the values of a and b?

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int tmp = a;
```

```
a = b;
```

```
b = a;
```

A. 0

B. 1

C. 2

D. 3

E. 4+

What does this function do?

```
public static void mystery(int[] arr) {  
    int minindex = 0;  
    for (int i=0; i<arr.length; i++) {  
        if (arr[i] < arr[minindex]) {  
            minindex = i;  
        }  
    }  
    int tmp = arr[0];  
    arr[0] = arr[minindex];  
    arr[minindex] = tmp;  
}
```

Swaps the minimum value with the value at index 0

```
public static void mystery(int[] arr) {  
    int minindex = 0;  
    for (int i=0; i<arr.length; i++) {  
        if (arr[i] < arr[minindex]) {  
            minindex = i;  
        }  
    }  
    int tmp = arr[0];  
    arr[0] = arr[minindex];  
    arr[minindex] = tmp;  
}
```

What does this code do?

```
static void sort(int arr[]) {  
    int n = arr.length;  
    for (int i = 0; i < n - 1; i++) {  
        int minIndex = i;  
        for (int j = i + 1; j < n; j++){  
            if (arr[j] < arr[minIndex]){  
                minIndex = j;  
            }  
        }  
        int temp = arr[minIndex];  
        arr[minIndex] = arr[i];  
        arr[i] = temp;  
    }  
}
```

Sorts the array! "Selection Sort"

```
static void sort(int arr[]) {  
    int n = arr.length;  
    // One by one move boundary of unsorted subarray  
    for (int i = 0; i < n - 1; i++) {  
        // Find the minimum element in unsorted array  
        int minIndex = i;  
        for (int j = i + 1; j < n; j++)  
            if (arr[j] < arr[minIndex])  
                minIndex = j;  
  
        // Swap the found minimum element with the first element  
        int temp = arr[minIndex];  
        arr[minIndex] = arr[i];  
        arr[i] = temp;  
    }  
}
```

Hungarian folk dance to sort an array



Bubble Sort

Plug into programiz

```
public class Bubble {  
    public static void main(String[] args) {  
        int[] arr = {5, 2, 8, 7, 1, 3, 9, 4, 6};  
        bubbleSort(arr);  
        for(int i = 0; i < arr.length; i++) {  
            System.out.print(arr[i] + " ");  
        }  
    }  
  
    public static void bubbleSort(int[] arr) {  
        int n = arr.length;  
        for (int i = 0; i < n - 1; i++) {  
            // every time the i loop executes,  
            // we move 1 more element to correct spot  
            // into the correct place at the end of the array  
            for (int j = 0; j < n - i - 1; j++) {  
                if (arr[j] > arr[j + 1]) {  
                    // swap if arr[j] > arr[j + 1]  
                    int temp = arr[j];  
                    arr[j] = arr[j + 1];  
                    arr[j + 1] = temp;  
                }  
            }  
        }  
    }  
}
```

Write out code for bubble sort

Basically, where we swap every value with the one next to it

Return an array from a function

What function declaration indicates that we are returning an array from a function?

- A. `public static int func(int x)`
- B. `public static int func(int[] x)`
- C. `public static int[] func(int x)`
- D. `public static int[] func(int[] x)`
- E. not exactly one of the above

Return an array from a function

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- C. `public static int[] func(int x)`
- D. `public static int[] func(int[] x)`
- E. not exactly one of the above**

Both of these are correct! What matters is the `int[]` of the return type! The parameter doesn't even matter

Write a function that return the min and the max in an array

```
public static int[] findmaxmin(int[] arr) {  
    int max=arr[0];  
    int min=arr[0];  
    for (int i=0; i<arr.length; i++) {  
        if (arr[i] > max) {  
            max=arr[i];  
        }  
        if (arr[i] < min) {  
            min=arr[i];  
        }  
    }  
  
    _____  
}
```

What do we put in the blank?

- A: return new int[min, max];
- B: return new int[2];
- C: return new int[] {min,max};
- D: return new int{min, max};
- E: return (min, max);

Write a function that return the min and the max in an array

```
public static int[] findmaxmin(int[] arr) {  
    int max=arr[0];  
    int min=arr[0];  
    for (int i=0; i<arr.length; i++) {  
        if (arr[i] > max) {  
            max=arr[i];  
        }  
        if (arr[i] < min) {  
            min=arr[i];  
        }  
    }  
    int [] bounds = new int[2];  
    bounds[0]=min;  
    bounds[1]=max;  
    return bounds; _____
```

What do we put in the blank?

A: return new int[min, max];

B: return new int[2];

C: return new int[] {min,max};

D: return new int{min, max};

E: return (min, max);

Why are the others wrong?

2D Arrays

So far all our lists are simple, 1D lists like
{ 12, 3, 5, 1, 9, 7, 22, 18, 42 }

But we've also seen Tables and 2D grids (e.g. SimpleGrid) that where elements in the grid are referenced using a *row* and a *column*.

Those are examples of two-dimensional arrays!

2D Arrays

Creating a 2D int array:

```
int [ ] [ ] my2dList = new int [10][10];
```

Accessing elements in the list:

```
thatElement = my2dList[2][5];
```

Otherwise, they're pretty much the same as the lists we've looked at.

DONE FOR NOW